

Course Contents:

Introduction; **Software Design:** Software architecture, Design patterns, OO analysis & Design, Design for re-use. **Using API's:** API programming, Class browsers and Related tools, Component based computing. **Software tools and Environment:** Requirements analysis and design modeling tools, Testing tools, Tool integration mechanisms.

Introduction**The Term Software**

Software is a set of instructions used to acquire inputs and to manipulate them to produce the desired output in terms of functions and performance as determined by the user of the software. It also includes a set of documents, such as the software manual, meant to help users understand the software system.

Software Tree:**Today's software is comprised of:**

1. **Source Code:** These are programs or codes written by the programmers using any particular programming language such as Java, VB, Python, etc.
2. **Executable:** These are compiled codes that can run without the source code and can be shared with other users to run independent of the source program.
3. **Design Documents:** These are different designs necessary when trying to develop software such as input design, output design, UML, use case design, class diagrams, software architecture, etc.
4. **Operations:** These are the functions expected of any software to be developed for any purpose by the software engineer of the developer. Functions can be listed to guide the developer on the way forward on each module of the software. It may be programmer centred, generic, or customized, etc.
5. **System Manuals:** Manuals are very important for the users, there is need for user's guide or user manual, technical manual, complete system manual, etc.

6. Installation: After the development of the software by the developer or programmer, it must be installed on the user's computer system. The process of making it function form the user's system is call installation which depends on the hardware and the operating system.
7. Implementation: After Installation, the use of the software for real life job by the organization is very important not to affect the existing system. This can be by any technique such as direct, pilot, phased or parallel technique depending on the type of the organisation, the software and the use.

Software also includes:

- i. Instructions (computer programs) that when executed provide desired functions and performance.
- ii. Data structures that enable the programs to adequately manipulate information.
- iii. Documents that describe the operation and use to dispatch the goods. The facilities and features could be optional and based on customer choices.

Software is developed keeping in mind certain hardware and operating system considerations, known as the platform. Hence, software is described along with its capabilities and the platform specifications that are required to run it.

Note: Practical on how to differentiate between source code and executable USING Visual Basic

The Term Engineering

Engineering generally means to make, achieve or get through contrivance or guile while contrivance means something, as a machine, devised for a particular function, something invented and specialised mechanical devices. Guile in its own case means deceit which is related to finding solution to problems that are not related to machines but cunning in nature or deceived in nature. More than a discipline or a body of knowledge, engineering is a verb, an action word, a way of approaching a problem. (Roger, 2005).

Adam et al (2001) stated that engineering means the profession and activity of designing the way, roads, bridges, machines etc. or machinery and equipment used or developed as a result of knowledge.

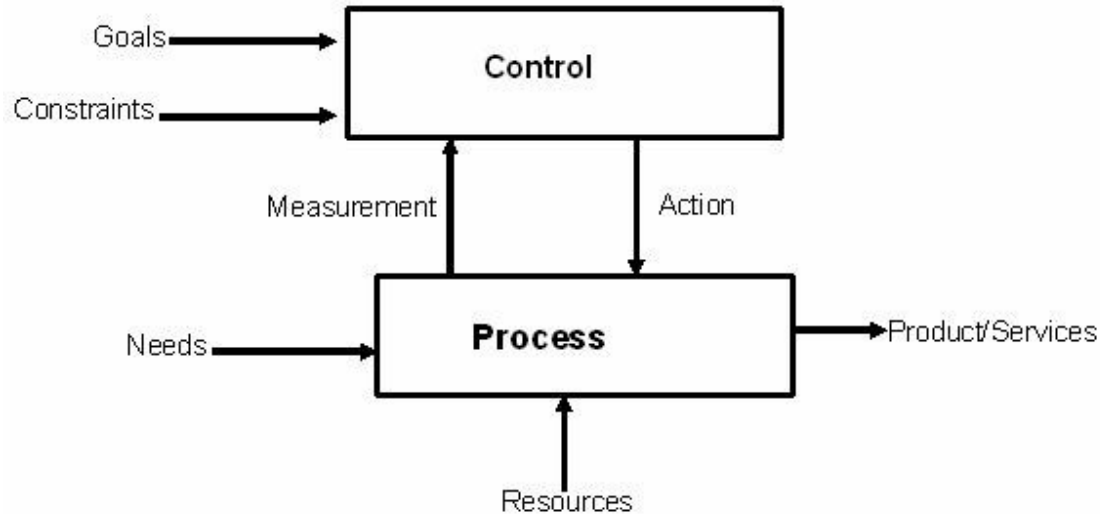


Fig 2.1: A model of Engineering

Source: Moore (1998). Software Engineering Standards, IEEE, 1998

The model above shows that, for any engineering process, there must be goals to be achieved while constraints will be associated with it which may likely affect one way or the other. Engineer is to control everything using measurement, action and process by getting the needs and necessary resources which will help in achieving the goals to produce the product or provide the services. This is a complete definition of engineering.

The Term Software Engineering

Software Engineering is commonly used for larger and complex software systems. It can be referred to as the application of engineering principles in software development. The emphasis on software engineering in computing is on how to achieve best practices and engineering processes that can be applied to software development.

Software engineering is the process of analyzing user needs and designing, constructing, and testing end user applications that will satisfy these needs through the use of **software** programming languages. It is the application of **engineering** principles to **software** development.

A few other important definitions given by several authors and institutions are as follows:

IEEE Comprehensive Definition. Software Engineering is the application of a systematic, disciplined, quantifiable approach to the development, operation and maintenance of software, i.e., the application of engineering to software.

Software Engineering deals with cost-effective solutions to practical problems by applying scientific knowledge in building software artifacts in the service of mankind.

OR

Software Engineering is the application of methods and scientific knowledge to create practical cost-effective solutions for the design, construction, operation and maintenance of software.

OR

Software Engineering is a discipline whose aim is the production of fault free software that satisfies the user's needs and that is delivered on time and within budget.

OR

The term Software Engineering refers to a movement, methods and techniques aimed at making software development more systematic.

Computer Science VS Software Engineering

- Computer Science is the study of how computers work, mostly from the theoretical and mathematical perspective.
- You should choose Computer Science if you like math, logic, or if you want to get into a specialized field in Computer Science such as artificial intelligence, machine learning, security, or graphics.

- Software Engineering is the study of how software systems are built, including topics such as project management, software architecture, risk management, quality assurance, software testing, etc.
- You should choose Software Engineering if you are more interested in the hands-on approach, and if you want to learn the overall life cycle of how software is built and maintained.
- Both Computer Science and Software Engineering teach fundamentals of programming and computer science, so you can choose either one to become a software developer.
- The core computer science requirements are similar as well, ranging over algorithms, data structures, and operating systems.

The key differences are:

- **Software Engineering** has more requirements in software engineering fundamentals, such as software testing, design, and software requirements specification.
- **Computer Science** allows more electives in higher-level computer science courses. You can choose from a wide range of topics such as security, software engineering fundamentals, computer vision, machine learning, and database management.

Importance of Software

Computer software has become a driving force.

1. It is the engine that drives business decision making.
2. It serves as the basis for modern scientific investigation and engineering problem-solving.
3. It is embedded in all kinds of systems, such as transportation, medical, telecommunications, military, industrial processes, entertainment, office products, etc.
4. It is important as it affects nearly every aspect of our lives and has become pervasive in our commerce, our culture, and our everyday activities. Software's impact on our society and culture is significant.

As software importance grows, the software community continually attempts to develop technologies that will make it easier, faster, and less expensive to build high-quality computer programs.